ADITYA SANJAY MALKAR

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PROFESSIONAL SUMMARY

MS Data Science candidate specializing in Agentic AI, NLP, and LLM optimization. Proven at architecting autonomous agents using AWS Bedrock and LangChain to automate complex workflows. Expert in building scalable ML pipelines and safety frameworks to ensure high-integrity model outputs.

EDUCATION

Stevens Institute of Technology – Hoboken, NJ, USA **Anticipated Graduation: May 2026**
Master of Science in Data Science | GPA – 3.7
Relevant Coursework: Deep Learning, NLP, Statistical Methods, Database Management (SQL), Time Series Analysis.

University of Mumbai – Mumbai, Maharashtra, India **Graduated: May 2024**
Bachelor of Engineering in Computer Engineering | CGPA - 8.37

EXPERIENCE

Prodigy Infotech, *(Machine Learning Intern- Image Classification)* – Mumbai, Maharashtra, India **Aug 2023 - Sep 2023**

- Collaborated on VisionNet, improving classification accuracy by 15% through hyperparameter tuning and feature selection.
- Optimized model inference for real-time classification using PyTorch, achieving a 25% reduction in latency by streamlining computation graphs.
- Enhanced data ingestion throughput by 35% by implementing a multi-threaded image preprocessing pipeline in Python, significantly accelerating the training lifecycle.

TechnoHacks Edutech, (*Machine Learning Intern- Model Optimization)* – Nashik, Maharashtra, India **Jul 2023 - Aug 2023**

- Implemented automated preprocessing pipelines (SMOTE), reducing data processing time by 30%.
- Built an AdaBoost-based anomaly detection model, reducing error rates by 20% through rigorous statistical validation.
- Optimized SQL data extraction processes for large-scale datasets, achieving a 45% improvement in query performance for downstream model training.

SKILLS

- **AI & Agentic Architectures:** Agentic AI (Goal-seeking, Tool-use), ReAct, Chain-of-Thought (CoT), RAG Pipelines, Multi-agent Orchestration.
- **Generative AI & LLMs:** LLMs (Claude V2, OpenAI API), LangChain, Sentiment Analysis, NER, Tokenization.
- **Programming & Databases:** Python (Pandas, Scikit-learn, PyTorch, TensorFlow), Java, C++, SQL (Data Extraction), R.
- **Cloud & Data Engineering:** AWS (SageMaker, Bedrock, Glue, EC2, RDS), Apache Spark, ETL/ELT Pipelines, Docker.
- **Machine Learning:** Supervised/Unsupervised Learning, Model Alignment & Safety, Outlier Detection, Statistical Validation.

ACADEMIC PROJECTS

AI-Powered Agentic Career Advisor (LLM Alignment & Reasoning), *MS Data Science* **Aug 2025 - Dec 2025**

- Architected an autonomous agent using AWS Bedrock and LangChain, implementing goal-seeking and reasoning capabilities.
- Designed a RAG pipeline that increased response relevance by 40% through integration with external career datasets.
- Developed a safety and alignment framework that reduced model hallucinations by 75%, ensuring high-reliability responses.

Multi-Agent System for Automated Licensing (Tool Orchestration), *MS Data Science* **May 2025 - Jul 2025**

- Engineered a dual-agent architecture using LangChain and OpenAI to automate complex hospital credentialing workflows.
- Developed a Parser Agent to extract 65+ key entities from unstructured PDFs into standardized JSON and a Form Filler Agent for automated field mapping.
- Deployed a scalable multi-tenant platform on AWS (EC2, RDS), reducing manual data entry time by approximately 90%.

Real-Time Speech-to-Speech Translation (Low-Latency AI Engineering), *MS Data Science* **Jan 2025 - Apr 2025**

- Architected a concurrent translation pipeline using PyTorch, Whisper, and MarianMT, achieving a pipeline depth of ~2.8 sec.
- Implemented a custom Voice Activity Detection (VAD) system with <3ms frame latency, reducing false positives to 1.8%.
- Engineered a hallucination filter to remove spurious audio artifacts and optimized processing for 200ms audio chunks.

MultiModal Twitter Sentiment Analysis (Deep Learning & Evaluation), *MS Data Science* **Aug 2024 - Dec 2024**

- Analyzed and cleaned 1.6 million unstructured tweets using tokenization, lemmatization, and emoji handling.
- Developed a deep learning pipeline (LSTM/Conv1D) achieving 85% accuracy in identifying novel sentiment signals.
- Conducted a comparative analysis between Bi-LSTM and BART models, achieving an 18% higher F1-score than traditional method

CERTIFICATIONS & LICENSES

- AWS Certified AI Practitioner, Amazon Web Services | Certificate
- AWS Certified Machine Learning Engineer Associate, Amazon Web Services | Certificate